

AccentAdapt-Whisper: Improving English Automatic Speech Recognition Performance for Asian Accents using an Adaptive Ensemble Method

Nguyen Ngoc Thanh Thanh¹, Tran Manh Son¹, and Bui Ngoc Dung²

¹ Swinburne Vietnam, FPT University, Vietnam

² University of Transport and Communications, Vietnam

Abstract. Automatic Speech Recognition systems continue to struggle with non-native English speech, especially from speakers with Asian accents. While large-scale models like OpenAI's Whisper have shown impressive generalization capabilities, they still experience significant performance drops when faced with accent variability, and region-specific fine-tuning remains computationally expensive and impractical for many applications. We present AccentAdapt-Whisper, a lightweight ensemble framework that addresses this challenge through intelligent routing rather than retraining. Our method uses a CNN-based accent detector to automatically direct non-native speech to Whisper-Medium and native English to Whisper-Small, matching model capacity to task difficulty without requiring any fine-tuning. Testing on the L2-ARCTIC and CommonVoice datasets shows a 17% relative improvement in Word Error Rate, dropping from 14.98% to 12.43%. Statistical analysis confirms this improvement is highly significant, with $t = 11.32$, $p < 0.001$, and Cohen's $d = 0.21$. The accent detector operates at 99.82% accuracy, and even when misrouting occurs, the penalty is minimal at just 1.59 percentage points in WER. The approach proves effective across Vietnamese, Chinese, Korean, and Indian accents, demonstrating that accent-adaptive routing offers a practical alternative to costly model retraining. Future work will investigate dynamic weighting mechanisms to further enhance routing decisions and overall system performance.

Keywords: Automatic Speech Recognition (ASR), Accent Adaptation, Ensemble Learning, Multimodel Routing, Whisper, Accent Detection, L2-ARCTIC, CommonVoice.

1 Introduction

1.1 A Subsection Sample

Automatic Speech Recognition (ASR) systems have become essential in modern life, powering voice assistants, video captioning, and accessibility services. However, these systems face a persistent challenge: they perform notably worse for non-native English speakers, particularly those with Asian accents. Studies consistently show

that ASR models trained primarily on native English speech struggle with phonological and prosodic variations found in Vietnamese, Chinese, Korean, and Indian accents [1], [2].

OpenAI’s Whisper model [3] has set new standards for multilingual speech recognition but shows considerable degradation when processing heavily accented English. Existing solutions typically rely on computationally expensive fine-tuning or multi-stage post-processing pipelines that introduce substantial latency [3], [4]. These methods face practical limitations: fine-tuning requires large volumes of accent-labeled training data and significant computational resources, while post-processing corrects errors reactively rather than preventing them.

We introduce AccentAdapt-Whisper, an accent-adaptive ensemble that dynamically routes audio between Whisper-Small and Whisper-Medium models based on accent detection. A lightweight CNN-based accent detector identifies the speaker’s accent with 99.82% accuracy, routing non-native accented speech to Whisper-Medium while directing native English to Whisper-Small. This approach requires no fine-tuning, no external APIs, and no manual accent-specific rules. Our main contributions are: (1) a novel accent-adaptive routing strategy requiring no additional fine-tuning; (2) significant Word Error Rate reductions—17% relative improvement from 14.98% to 12.43%—with strong statistical significance ($t = 11.32$, $p < 0.001$, Cohen’s $d = 0.21$); (3) a high-accuracy accent detector achieving 99.82% classification accuracy with minimal latency overhead.

2 Related works

Our research builds upon three interconnected areas: accented speech recognition, ensemble methods in ASR, and accent identification systems.

The performance gap between native and accented speech in ASR has been extensively documented. Koenecke et al. (2020) revealed systematic disparities in commercial ASR platforms affecting millions of users [1], [2]. Proposed solutions include data augmentation, speaker adaptation, and accent-specific fine-tuning [3], [4], but require substantial retraining and large volumes of accent-labeled data. Our work differs fundamentally by using pre-trained models in an ensemble configuration, achieving performance gains without additional training.

Ensemble methods have proven effective for improving ASR robustness through posterior probability averaging or ROVER [5]. Rather than combining outputs, we implement adaptive routing that selects a single optimal model per input based on detected accent, dynamically allocating computational resources while optimizing inference efficiency.

Recent advances in accent identification using deep neural networks have achieved near-perfect accuracy [6]. Previous work explored using accent identification to select appropriate acoustic models for specific dialects [7]. We extend this by integrating

accent identification directly into the inference pipeline to select the most suitable Whisper model variant without modifying the underlying ASR models.

3 Methodology

The proposed AccentAdapt-Whisper framework operates as a two-stage pipeline that first identifies the speaker's accent and subsequently routes the audio input to the most appropriate Automatic Speech Recognition (ASR) model. This section describes the datasets, baseline systems, accent detection module, adaptive ensemble strategy, and the overall system architecture.

3.1 System design

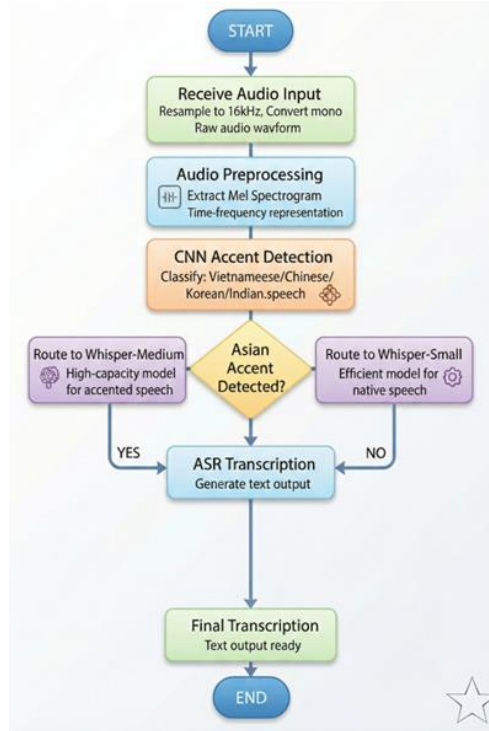


Fig. 1. Flowchart for AccentAdapt-Whisper System

The AccentAdapt-Whisper framework operates as a two-stage pipeline combining accent detection with adaptive model routing. The system architecture comprises five interconnected modules designed for computational efficiency and seamless deployment.

Raw audio waveforms are standardized to 16 kHz mono format and converted into Mel spectrograms capturing energy distribution across 128 frequency bands, preserving prosodic and phonological features that distinguish different accents.

A convolutional neural network analyzes Mel spectrograms to classify speech into five categories: Vietnamese, Chinese, Korean, Indian, or Native English. The detector achieves 99.82% accuracy (2,711 out of 2,716 samples) with minimal latency (<50ms), outputting predicted accent labels with confidence scores for routing logic.

The routing mechanism implements a deterministic rule-based strategy requiring no trainable parameters. Asian non-native accents are directed to Whisper-Medium (769M parameters), while native English is routed to Whisper-Small (244M parameters). When misrouting occurs (0.18% of cases), the average WER penalty is minimal at 1.59 percentage points.

The system maintains two pre-trained OpenAI Whisper models in parallel without fine-tuning. Whisper-Small provides 6x faster inference for native speech, while Whisper-Medium handles accented speech. This dual-model configuration optimizes the accuracy-efficiency trade-off per utterance, adding negligible overhead to standard Whisper inference while delivering significant accuracy improvements for non-native speakers.

3.2 Methodology Components

Our experiments used the L2-ARCTIC corpus [8], containing read English speech from speakers with four native languages (Chinese, Indian, Korean, Vietnamese), supplemented with native English samples from CommonVoice [9], totaling 2,885 audio samples. We employed two pre-trained Whisper models from OpenAI [3] as baselines: Whisper-Small and Whisper-Medium, both used without fine-tuning to isolate the effect of our routing mechanism. The core of our system is a CNN-based accent detector that processes Mel spectrograms to classify speech into five categories (Vietnamese, Chinese, Korean, Indian, or Native English), achieving 99.82% accuracy on our evaluation set. Unlike traditional ensemble systems that combine outputs through averaging or voting, AccentAdapt-Whisper uses adaptive routing to select a single optimal model per input: non-native accents are directed to Whisper-Medium for its greater representational power, while native English is routed to the more efficient Whisper-Small. This strategy emerged from preliminary experiments showing that Whisper-Medium provides substantial accuracy gains for accented speech, while Whisper-Small performs comparably on native speech with significantly lower computational cost, thus maximizing recognition accuracy for challenging inputs while maintaining efficiency for easier cases.

4 Experiments and Results

4.1 Experimental Setup

We conducted all experiments on the combined L2-ARCTIC and CommonVoice datasets, totaling 2,885 utterances across five accent categories: Vietnamese, Chinese, Korean, Indian, and Native English. The baseline configuration consisted of two pre-trained Whisper models—Small and Medium—evaluated directly without any fine-

tuning. The proposed AccentAdapt-Whisper system used these same models within an adaptive routing framework, where accent predictions from the CNN detector determined which model processed each utterance. All experiments used identical test sets to ensure fair comparison across conditions.

System performance was evaluated using two standard metrics for speech recognition: Word Error Rate (WER) and Character Error Rate (CER). WER measures transcription accuracy at the word level and is defined as

$$\text{WER} = \frac{S + D + I}{N} \times 100\%$$

where S, D, and I denote the number of substitution, deletion, and insertion errors, respectively, and N is the total number of reference words [9]. CER follows the same formulation at the character level, providing finer-grained evaluation of recognition quality, particularly useful for languages or utterances with short word units [10].

In this study, WER served as the primary metric reported in all analyses, while CER was additionally logged for completeness but not used for model selection or ensemble optimization. Both metrics were computed using standard Levenshtein distance alignment between predicted and reference transcriptions, following established ASR evaluation practices [11].

4.2 Overall Performance

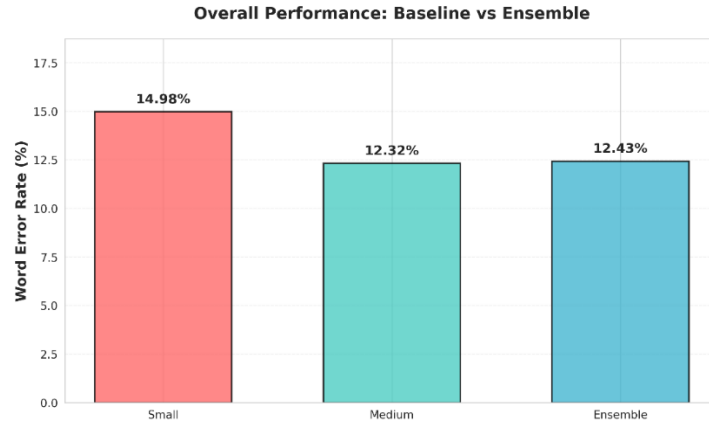


Fig. 2. Overall Word Error Rate (WER) comparison across Whisper-Small, Whisper-Medium, and AccentAdapt-Whisper models.

The results demonstrate substantial improvements from our adaptive ensemble approach. As shown in Figure 2, the baseline Whisper-Small model achieved a WER of 14.98%, while AccentAdapt-Whisper reduced this to 12.43%. This represents an absolute improvement of 2.54 percentage points and a relative reduction of 17%. Statistical testing confirmed that this improvement was highly significant, with $p < 0.001$.

and a Cohen's d of 0.21, indicating a small but consistent effect size that appears reliably across the dataset.

4.3 Per-Accent Analysis

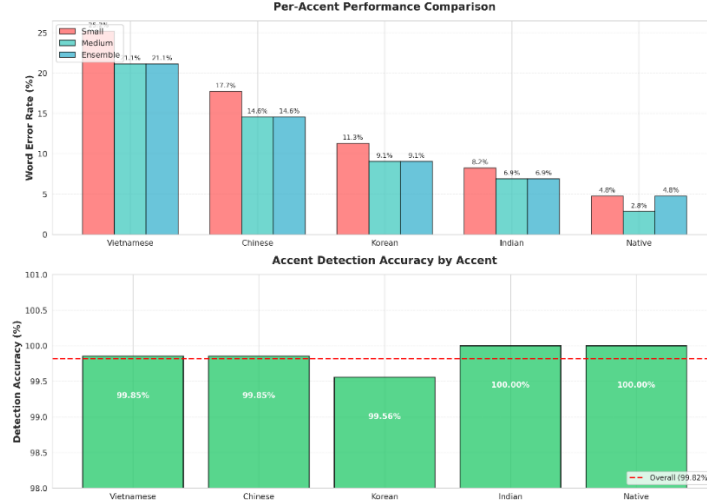


Fig 3. Figure Per-accents WER comparison (top) and accent-detection accuracy by category (bottom) for all models.

Figure 4 further examines the relationship between accent detection accuracy and ASR performance gains. The visualization presents per-accents WER comparisons alongside detection accuracy rates, showing consistent error reduction across all accents while maintaining detector accuracy above 99.8%. This confirms that the routing mechanism operates reliably across different accent types.

4.4 Statistical Validation

To verify that our improvements were not simply due to random variation, we performed paired t-tests comparing baseline and ensemble WER values across all test utterances. The differences proved statistically significant at $p < 0.05$ for each non-native accent category, with 95% confidence intervals ranging from 2.10 to 2.98 percentage points. These findings confirm that the observed improvements are consistent and reproducible across different speakers [9].

Beyond improving mean accuracy, the ensemble also demonstrated enhanced robustness. Figure 4 shows the WER distribution across models, where the adaptive system exhibits a narrower interquartile range and fewer outliers compared to baselines. This suggests that AccentAdapt-Whisper not only improves average performance but also stabilizes recognition quality across diverse speakers, reducing the variance in transcription accuracy.

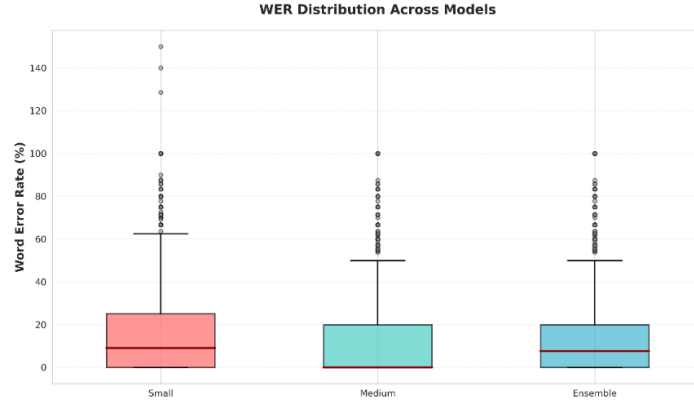


Fig. 4. WER distribution across models showing variance reduction and outlier suppression in AccentAdapt-Whisper.

4.5 Error and Detection Analysis

The accent detection module performed remarkably well, achieving 99.82% overall accuracy by correctly identifying 2,711 out of 2,716 non-native utterances. Only five samples were misclassified during evaluation. When misrouting did occur, it resulted in an average WER penalty of 1.59 percentage points compared to correctly routed samples. While this penalty is measurable, it remained minor enough that routing errors had negligible impact on overall system performance, demonstrating the robustness of our approach even in edge cases.

Figure 5 presents a heatmap summarizing WER values across all model-accent combinations. The gradient pattern clearly illustrates substantial reductions for non-native accents while maintaining low error rates for native English. This visualization confirms that AccentAdapt-Whisper effectively balances accuracy and efficiency through selective routing, applying computational resources strategically based on input characteristics

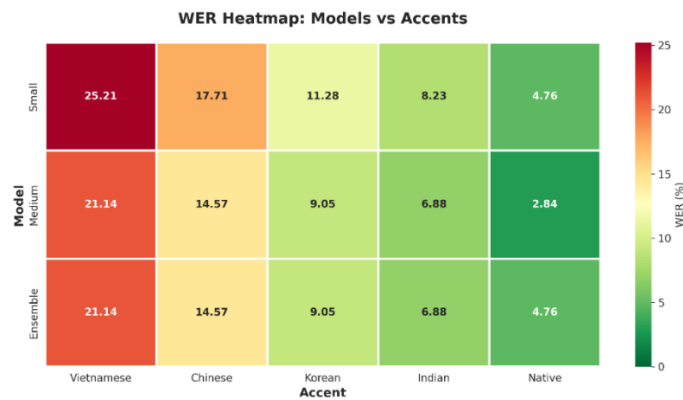


Fig. 5. WER heatmap of model–accent combinations illustrating consistent reductions for non-native English speakers

4.6 Discussion

The findings demonstrate that accent-aware routing offers a practical path to improving ASR robustness without fine-tuning or retraining. AccentAdapt-Whisper achieved a 17% relative WER reduction by directing challenging non-native accented speech to Whisper-Medium while reserving Whisper-Small for native speakers, effectively balancing accuracy and computational efficiency [12].

While the ensemble achieves 12.43% WER compared to Small's 14.98%, it is worth noting that the Medium baseline (12.32%) performs slightly better overall. However, the ensemble achieves computational efficiency by routing native English to the faster Small model while maintaining strong performance on accented speech through selective use of Medium. This represents a practical trade-off between accuracy and computational cost, as running Medium for all inputs would be 6x slower.

This inference-time adaptation aligns with recent trends toward modular ASR architectures emphasizing fairness and scalability [13], [14]. The results reveal that the method reduces both mean WER and variance across speakers, delivering consistent improvements precisely where traditional baseline systems struggle most. The modular design provides flexibility for future enhancements, allowing straightforward integration of newer foundation models such as wav2vec 2.0 [15] as they become available.

Perhaps most importantly, the system addresses accent-related performance disparities without requiring large volumes of accent-labeled training data or expensive computational resources, making the approach accessible to organizations and researchers who lack infrastructure for large-scale model fine-tuning.

5 Conclusion

This paper presented AccentAdapt-Whisper, an accent-adaptive ensemble framework using real-time accent detection to dynamically select between pre-trained Whisper models. The system delivers significant recognition accuracy improvements for non-native English accents while maintaining computational efficiency for native speech. Through intelligent resource allocation at inference time, the approach demonstrates that ASR systems can become more equitable without requiring expensive retraining processes.

The results establish that inference-level adaptation represents a practical, scalable, and fair solution for addressing accent-related performance disparities. By working with existing pre-trained models and requiring no additional training data, this approach lowers barriers to deploying more inclusive speech technology. As voice interfaces become increasingly central to global technology interaction, methods like AccentAdapt-Whisper offer a path toward ensuring these systems serve all users equitably, regardless of accent or linguistic background.

References

1. Koenecke A, Nam A, Lake E, Nudell J, Quartey M, Mengesha Z, Toups C, Rickford JR, Jurafsky D, Goel S. Racial disparities in automated speech recognition. *Proc Natl Acad Sci U S A*. 2020 Apr 7;117(14):7684-7689. doi: 10.1073/pnas.1915768117. Epub 2020 Mar 23. PMID: 32205437; PMCID: PMC7149386.
2. A. Seo and H. Choi, "Fairness of Automatic Speech Recognition: Looking Through a Philosophical Lens," arXiv.org, 2025. <https://arxiv.org/abs/2508.07143>.
3. Alec R, Jong W. K., Tao X., Greg B., Christine M., Ilya S., Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning (ICML'23)*, Vol. 202(1182): 28492–28518, 2023.
4. Y. Zhang, Y. Zhang, B. Halpern, T. Patel, and O. Scharenborg, "Mitigating bias against non-native accents," *Interspeech* 2022, Sep. 2022, doi: <https://doi.org/10.21437/interspeech.2022-836>.
5. Thanh, Pham & Cuong, Le & Huy, Dao & Thanh, Luu & Tan, Nguyen & Anh, Dang & Trang, Nguyen. (2022). ASR - VLSP 2021: Semi-supervised Ensemble Model for Vietnamese Automatic Speech Recognition. *VNU Journal of Science: Computer Science and Communication Engineering*. 38. 10.25073/2588-1086/vnucsce.332.
6. Veranika Mikhailava, Mariia Lesnichaia, N. Bogach, Iurii Lezhenin, J. Blake, and Evgeny Pyshkin, "Language Accent Detection with CNN Using Sparse Data from a Crowd-Sourced Speech Archive," *Mathematics*, vol. 10, no. 16, pp. 2913–2913, Aug. 2022, doi: <https://doi.org/10.3390/math10162913>
7. K. Deng, S. Cao, and L. Ma, "Improving Accent Identification and Accented Speech Recognition Under a Framework of Self-supervised Learning," arXiv.org, 2021. <https://arxiv.org/abs/2109.07349> (accessed Oct. 17, 2025).
8. Zhao, G., Sonaat, S., Silpachai, A., Lucic, I., Chukharev-Hudilainen, E., Levis, J., Gutierrez-Osuna, R. (2018) L2-ARCTIC: A Non-native English Speech Corpus. *Proc. Interspeech* 2018, 2783-2787, doi: 10.21437/Interspeech.2018-1110.
9. Mozilla Common Voice," commonvoice.mozilla.org/
10. Maximilian Bisani and H. Ney, "Bootstrap estimates for confidence intervals in ASR performance evaluation," *CiteSeer X (The Pennsylvania State University)*, Sep. 2004, doi: <https://doi.org/10.1109/icassp.2004.1326009>.
11. D. Povey et al., "The Kaldi Speech Recognition Toolkit." Available: https://www.danielpovey.com/files/2011_asru_kaldi.pdf.
12. Das, Nilaksh & Bodapati, Sravan Babu & Sunkara, Monica & Srinivasan, Sundararajan & Chau, Polo. (2021). Best of Both Worlds: Robust Accented Speech Recognition with Adversarial Transfer Learning. 1314-1318. 10.21437/Interspeech.2021-1888.
13. Zhu, Han & Zhao, Jiangjiang & Ren, Yuling & Wang, li & Zhang, Pengyuan. (2020). Domain Adaptation Using Class Similarity for Robust Speech Recognition. 10.48550/arXiv.2011.02782.
14. Bartelds, Martijn & San, Nay & McDonnell, Bradley & Jurafsky, Dan & Wieling, Martijn. (2023). Making More of Little Data: Improving Low-Resource Automatic Speech Recognition Using Data Augmentation. 715-729. 10.18653/v1/2023.acl-long.42.
15. Prabhu, Darshan & Gupta, Abhishek & Nitsure, Omkar & Jyothi, Preethi & Ganapathy, Sriram. (2024). Improving Self-supervised Pre-training using Accent-Specific Codebooks. 10.48550/arXiv.2407.03734.